

ALFAA39119

## 2-Butanone

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 2-丁酮, ACS  
**Product Description:** 2-Butanone

**Cat No. :** 39119  
**Synonyms** Methyl ethyl ketone; MEK; Ethyl methyl ketone  
**CAS No** 78-93-3  
**Molecular Formula** C4 H8 O

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
Characteristic - sweet

#### Emergency Overview

Highly flammable liquid and vapor. Causes serious eye irritation. May cause drowsiness and dizziness. May be harmful if swallowed. Repeated exposure may cause skin dryness or cracking. Hygroscopic.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Flammable liquids.                                 | Category 2 |
| Acute Oral Toxicity                                | Category 5 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements



**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation  
H336 - May cause drowsiness or dizziness  
H303 - May be harmful if swallowed

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion. Highly flammable. Hygroscopic.

**Health Hazards**

Causes serious eye irritation. May cause drowsiness or dizziness. May be harmful if swallowed.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its volatility. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces.

Contains a substance on the National Authorities Endocrine Disruptor Lists. Contains a known or suspected endocrine disruptor.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component           | CAS No  | Weight % |
|---------------------|---------|----------|
| Methyl ethyl ketone | 78-93-3 | <=100    |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention if symptoms occur. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Most important symptoms and effects**

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

**Specific Hazards Arising from the Chemical**

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation.

**Environmental Precautions**

Avoid release to the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Use spark-proof tools and explosion-proof equipment. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

**Specific Use(s)**

## 2-Butanone

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

## Control Parameters

| Component           | China                                                     | Taiwan                                     | Thailand     | Hong Kong                                                                                  |
|---------------------|-----------------------------------------------------------|--------------------------------------------|--------------|--------------------------------------------------------------------------------------------|
| Methyl ethyl ketone | TWA: 300 mg/m <sup>3</sup><br>STEL: 600 mg/m <sup>3</sup> | TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | TWA: 200 ppm | TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup><br>STEL: 300 ppm<br>STEL: 885 mg/m <sup>3</sup> |

| Component           | ACGIH TLV                     | OSHA PEL                                                                                                                                                                         | NIOSH                                                                                                        | The United Kingdom                                                                                                      | European Union                                                                                                       |
|---------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Methyl ethyl ketone | TWA: 200 ppm<br>STEL: 300 ppm | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 590 mg/m <sup>3</sup><br>(Vacated) STEL: 300 ppm<br>(Vacated) STEL: 885 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | IDLH: 3000 ppm<br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup><br>STEL: 300 ppm<br>STEL: 885 mg/m <sup>3</sup> | STEL: 300 ppm 15 min<br>STEL: 899 mg/m <sup>3</sup> 15 min<br>TWA: 200 ppm 8 hr<br>TWA: 600 mg/m <sup>3</sup> 8 hr Skin | TWA: 200 ppm (8h)<br>TWA: 600 mg/m <sup>3</sup> (8h)<br>STEL: 300 ppm (15min)<br>STEL: 900 mg/m <sup>3</sup> (15min) |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Exposure Controls

## Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard       | Glove comments                                                                                                               |
|----------------|-------------------|-----------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| Butyl rubber   | < 60 minutes      | 0.5 mm          | Level 4<br>EN 374 | Permeation rate 36 µg/cm <sup>2</sup> /min<br>As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

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|                                        |                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.                                                                                                                                              |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Type A Organic gases and vapours filter Brown conforming to EN14387    |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141 |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                        |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                     |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                                 |                                             |
|------------------------------------------------|-------------------------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Colorless                                       |                                             |
| <b>Physical State</b>                          | Liquid                                          |                                             |
| <b>Odor</b>                                    | Characteristic - sweet                          |                                             |
| <b>Odor Threshold</b>                          | No data available                               |                                             |
| <b>pH</b>                                      | No information available                        |                                             |
| <b>Melting Point/Range</b>                     | -87 °C / -124.6 °F                              |                                             |
| <b>Softening Point</b>                         | No data available                               |                                             |
| <b>Boiling Point/Range</b>                     | 80 °C / 176 °F                                  |                                             |
| <b>Flash Point</b>                             | -7 °C / 19.4 °F                                 | <b>Method -</b> CC (closed cup)             |
| <b>Evaporation Rate</b>                        | 3.7                                             | (Butyl Acetate = 1.0)                       |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.8 Vol%<br><b>Upper</b> 11.5 Vol% |                                             |
| <b>Vapor Pressure</b>                          | 105 mbar @ 20 °C                                |                                             |
| <b>Vapor Density</b>                           | 2.41                                            | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.806                                           |                                             |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                      |
| <b>Water Solubility</b>                        | 290 g/L (20°C)                                  |                                             |
| <b>Solubility in other solvents</b>            | No information available                        |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                             |
| Methyl ethyl ketone                            | 0.29                                            |                                             |
| <b>Autoignition Temperature</b>                | 404 °C / 759.2 °F                               |                                             |
| <b>Decomposition Temperature</b>               | No data available                               |                                             |
| <b>Viscosity</b>                               | 0.42 mPa.s @ 15°C                               |                                             |
| <b>Explosive Properties</b>                    | Not explosive                                   | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | Not oxidising                                   |                                             |
| <b>Molecular Formula</b>                       | C4 H8 O                                         |                                             |
| <b>Molecular Weight</b>                        | 72.11                                           |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                            |                               |
|----------------------------|-------------------------------|
| <b>Stability</b>           | Hygroscopic.                  |
| <b>Hazardous Reactions</b> | None under normal processing. |

## 2-Butanone

|                                 |                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                                                                                              |
| <b>Conditions to Avoid</b>      | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water. |
| <b>Materials to avoid</b>       | Strong oxidizing agents. Strong acids. Strong bases. Strong reducing agents. Ammonia. copper. Amines.                                 |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component           | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation              |
|---------------------|---------------------------|------------------------------|------------------------------|
| Methyl ethyl ketone | LD50 = 2483 mg/kg ( Rat ) | LD50 = 5000 mg/kg ( Rabbit ) | LC50 = 11700 ppm ( Rat ) 4 h |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met  
Not mutagenic in AMES Test

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Category 3

Results / Target organs

Central nervous system (CNS)

**(i) STOT-repeated exposure;** Based on available data, the classification criteria are not met

Target Organs

None known.

**(j) aspiration hazard;** Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting:  
Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**SECTION 12. ECOLOGICAL INFORMATION**

**Ecotoxicity effects**

**SAFETY DATA SHEET****2-Butanone**

| Component           | Freshwater Fish                            | Water Flea                                                                                                                                        | Freshwater Algae | Microtox                                                |
|---------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------|
| Methyl ethyl ketone | Lepomis macrochirus:<br>LC50=3,22 g/L 96 h | EC50: = 5091 mg/L, 48h<br>(Daphnia magna)<br>EC50: 4025 - 6440<br>mg/L, 48h Static<br>(Daphnia magna)<br>EC50: > 520 mg/L, 48h<br>(Daphnia magna) |                  | EC50 = 3403 mg/L 30<br>min<br>EC50 = 3426 mg/L 5<br>min |

**Persistence and Degradability**

Readily biodegradable

**Persistence**

Persistence is unlikely, based on information available.

| Component                                | Degradability |
|------------------------------------------|---------------|
| Methyl ethyl ketone<br>78-93-3 ( <=100 ) | 98% (28d)     |

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Methyl ethyl ketone | 0.29    | No data available             |

**Mobility in soil**

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport****UN-No**

UN1193

**Proper Shipping Name**

Ethyl methyl ketone (Methyl ethyl ketone)

**Hazard Class**

3

**Packing Group**

II

**IMDG/IMO****UN-No**

UN1193

**Proper Shipping Name**

Ethyl methyl ketone (Methyl ethyl ketone)

**Hazard Class**

3

**Packing Group**

II

**IATA**

## 2-Butanone

UN-No UN1193  
 Proper Shipping Name Methyl ethyl ketone  
 Hazard Class 3  
 Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component           | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|---------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Methyl ethyl ketone | X                                                   | X                                       | X    | X     | 201-159-0 | X    | X   | X     | X    | X    | X    | KE-24094 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
 Creation Date 13-Apr-2009  
 Revision Date 27-Apr-2024  
 Revision Summary New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

## 2-Butanone

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**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**